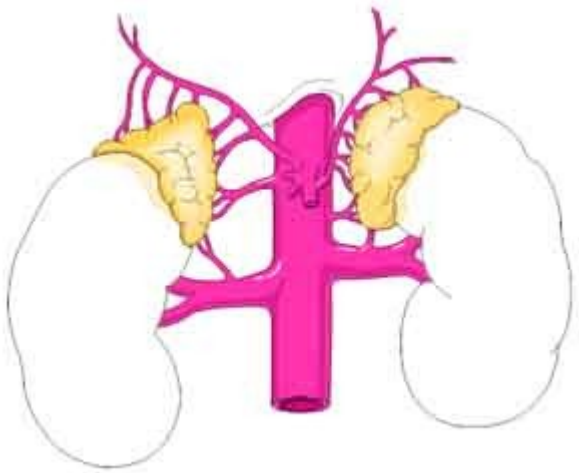
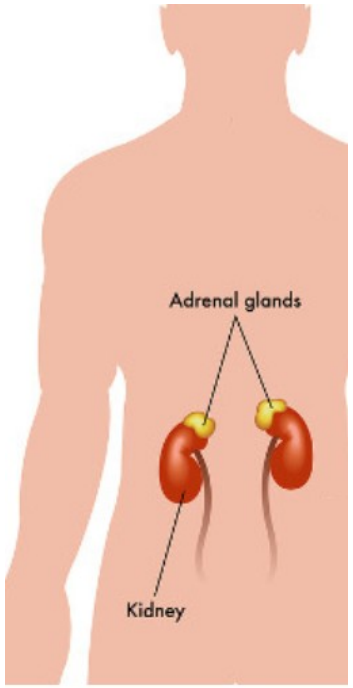
The Adrenal Glands

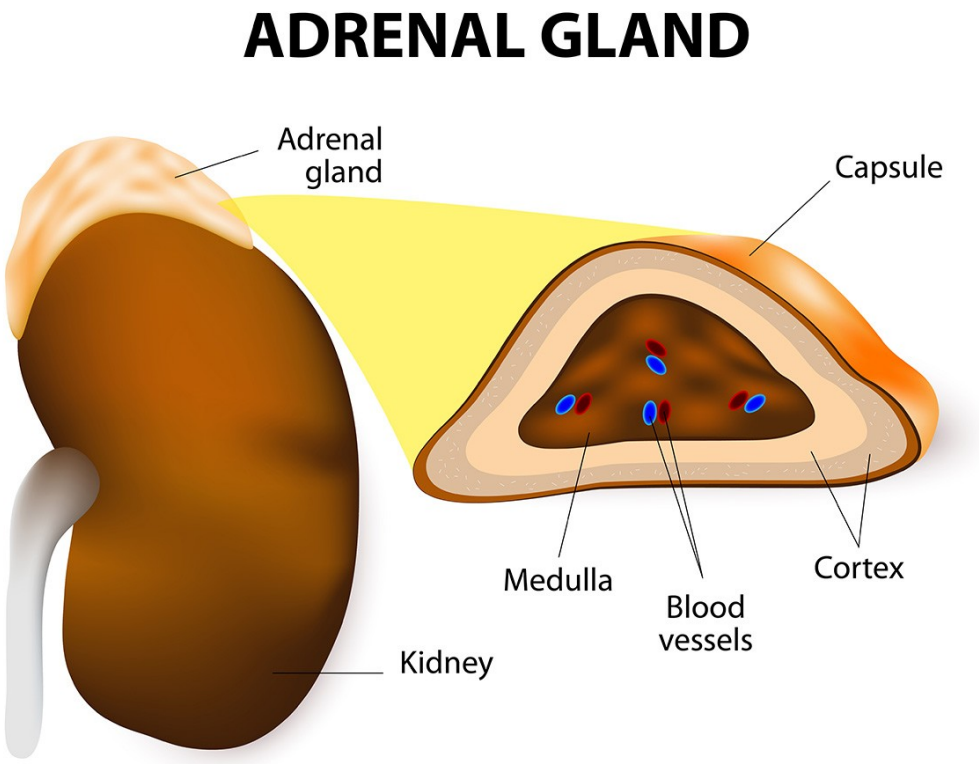


Structure, Hormones, Regulation and Disease

Adrenal Gland Location and Structure

- Located on top of each kidney
- Cortex – outer layer
- Medulla – inner layer

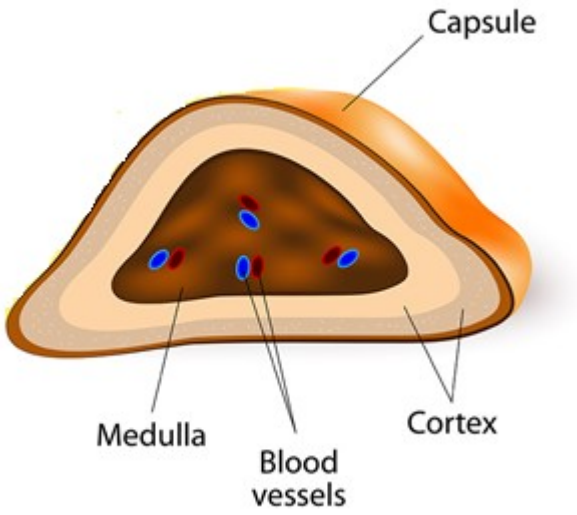




Learning Aim: Describe the location and structure of the adrenal glands

Adrenal Gland Hormones

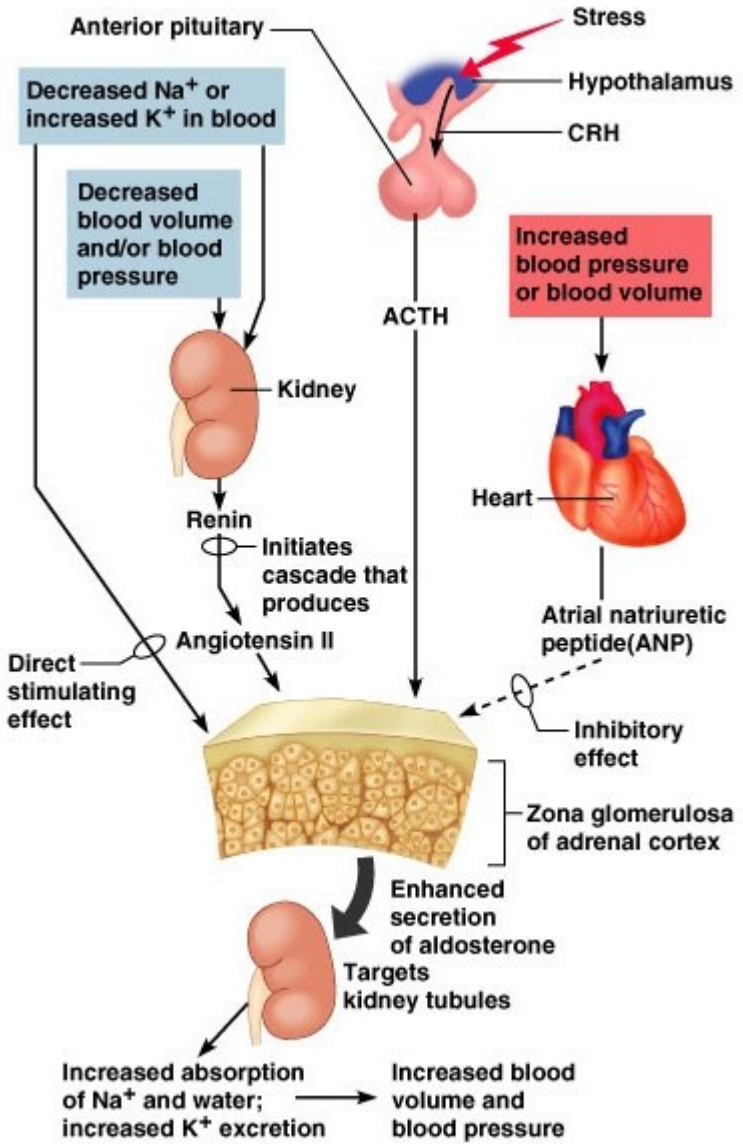
Adrenal Cortex (outside)	Adrenal Medulla (inside)
Aldosterone: • A mineralocorticoid • Role in maintaining salt and water levels in body, and therefore blood pressure • Acts on collecting ducts in kidney, causing increased Na^+ and water reabsorption and increased K^+ secretion.	**Catecholamines eg:** • Adrenaline, Noradrenaline, and some Dopamine • Involved in response to acute stress – "fight or flight. • Increase blood pressure, heart rate, respiratory function. • Direct blood away from digestive tract
Cortisol: • A glucocorticoid • Role in response to stress, illness and metabolism • Stimulates glucose production via gluconeogenesis and conversion of glycogen to glucose. • Also has potent anti-inflammatory effects	
Adrenal Androgens • Weak effects • Involved in male sex organ development in childhood	



Learning Aim: List, and describe the effects of hormones produced by the adrenal cortex and the adrenal medulla.

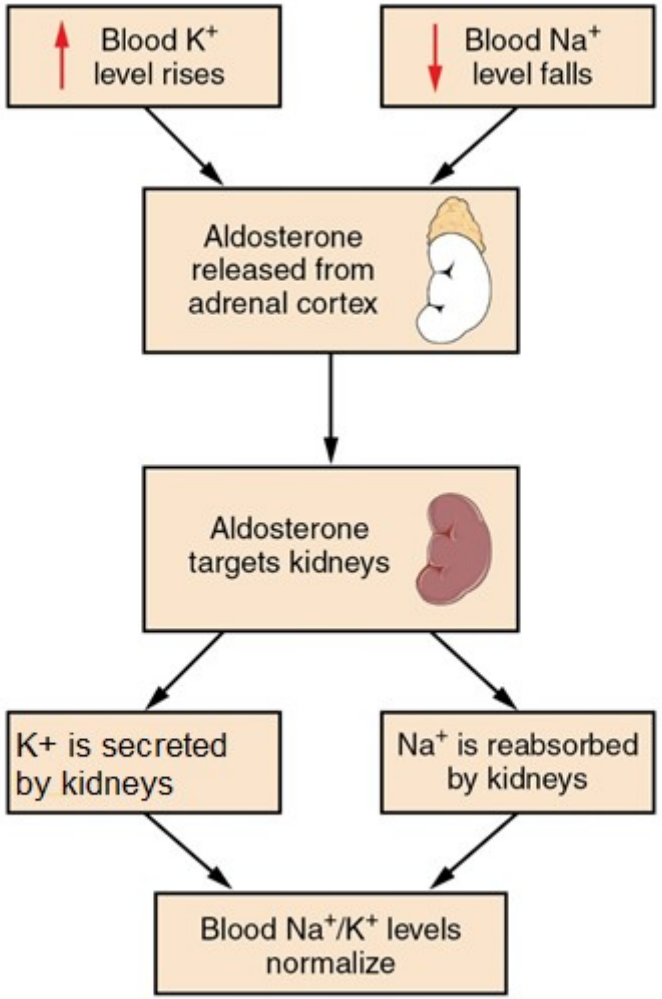
Regulation of Aldosterone release

- It's complicated!
- **For Year 12 ATAR Human Biology, focus on:**



Aldosterone:

- Mineralocorticoid
- Role in maintaining salt and water levels in body, and therefore blood pressure.
- Reliant on signalling from receptors in kidney and heart.
- Acts on collecting ducts in kidney, causing increased Na^+ and water reabsorption and increased K^+ secretion.

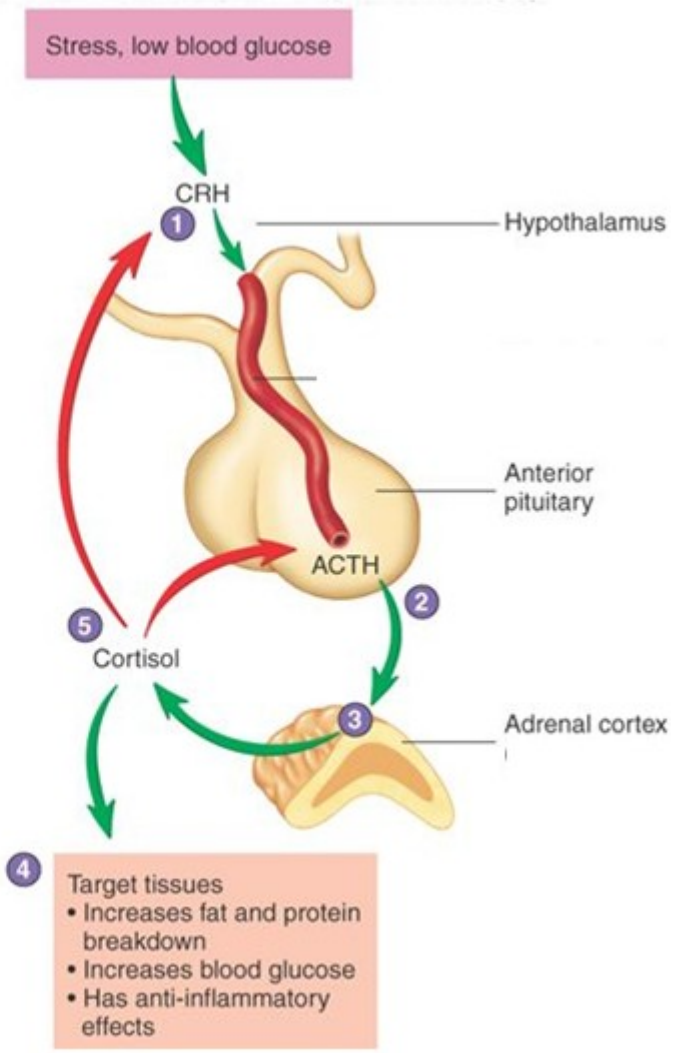


Learning Aim: Describe in detail the regulation and effects of aldosterone.

Regulation of Cortisol Secretion

1. Cortiocotropin-releasing hormone (CRH) is released from hypothalamic neurons in response to stress or low blood glucose and passes, by way of the hypothalamohypophyseal portal system, to the anterior pituitary.
2. CRH stimulates the secretion of adrenocorticotropic hormone (ACTH) from the anterior pituitary.
3. ACTH stimulates the secretion of cortisol from the adrenal cortex.
4. Cortisol acts on target tissues, resulting in increased fat and protein breakdown, increased glucose levels, and anti-inflammatory effects.
5. Cortisol has a negative-feedback effect because it inhibits CRH release from the hypothalamus and ACTH secretion from the anterior pituitary.





Hypothalamus

Detects:
- *high levels of stress
- * low blood glucose

Releases:
Corticotropin Releasing Factor (CRF)

Corticotropin Releasing Factor (CRF) *travels through the blood vessels of the infundibulum to the Anterior Pituitary*

Anterior Pituitary

CRF stimulates the glandular cells of the Anterior Pituitary to produce Adrenocorticotropic Hormone **(ACTH)**

ACTH *is released from the Anterior Pituitary and travels through the systemic circulation*

Adrenal Cortex

The ACTH binds to receptors in the Adrenal Cortex, stimulating the production of **Cortisol**

Cortisol *is released from the Adrenal Cortex and travels through the systemic circulation*

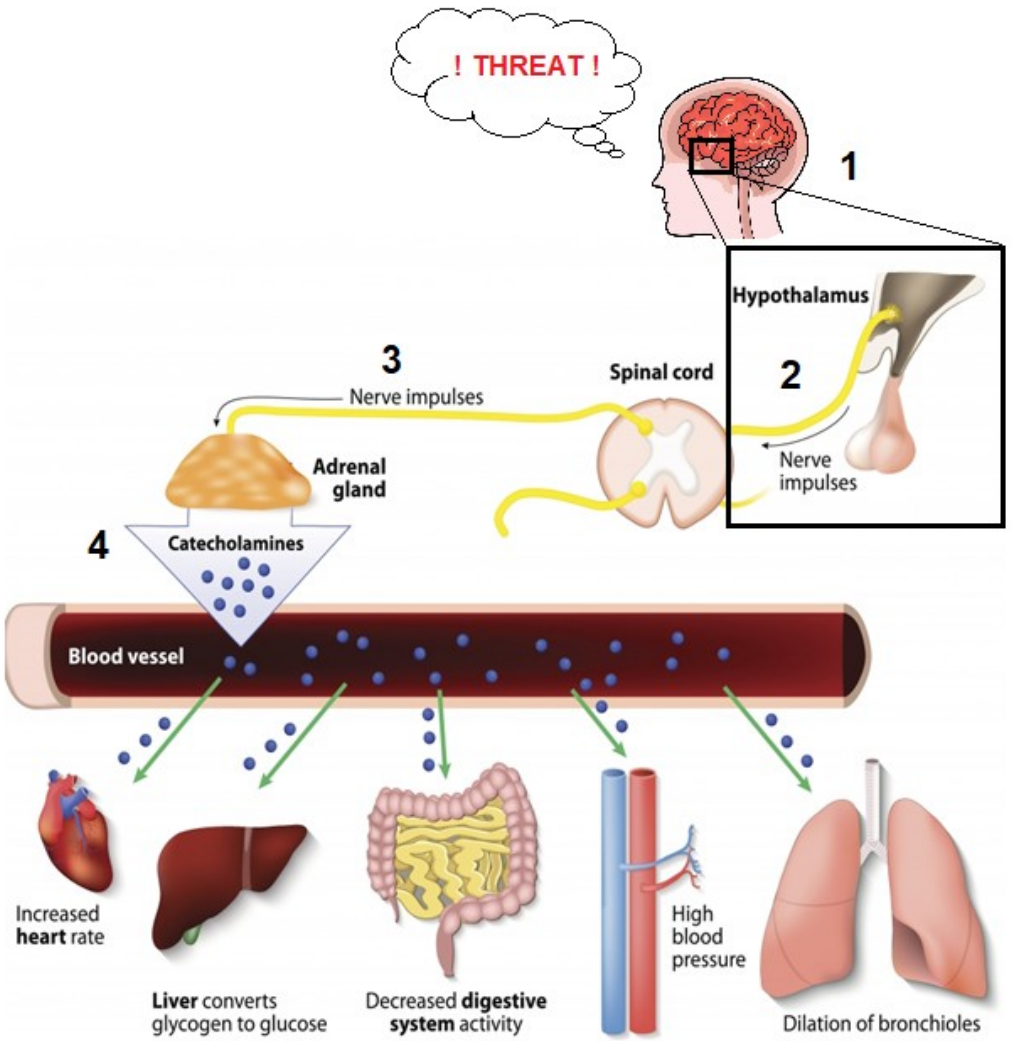
Body Tissues

- * increase fat and protein breakdown
- *increase blood glucose
- *produce anti-inflammatory effects

Rising levels of Cortisol inihit further production of CRF and ACTH, so that Cortisol levels stabilise. This is **Negative Feedback**

Learning Aim: Describe in detail the regulation and effects of cortisol.

Regulation of Catecholamine release



Interaction between nervous system and endocrine system.

1: Brain detects threat, and sends nerve signals to the hypothalamus

2: The hypothalamus sends nerve signals to the spinal cord (via the autonomic nervous system)

3: The spinal cord sends signals to the adrenal medulla

4: The adrenal gland produces catecholamines such as adrenalin and noradrenalin, producing a "fight or flight" response:

- Increased heart rate and blood pressure
- Dilation of bronchioles
- Liver converts glycogen to glucose, increasing blood glucose
- Digestive system activity decreases

5: When the threat stimulus is no longer detected, the hypothalamus will stop sending nerve impulses to the adrenal medulla.

Learning Aim: Describe in detail the regulation and effects of catecholamine release.

Disorders of the Adrenal Gland

- Are complex in symptoms, diagnosis and treatment
- Are not a focus in the curriculum